

Annotation Questionnaire

Annotation Instructions

For each of the eighteen dialog qualities, the detailed instructions and examples are shown below. These instructions were available for the worker to expand for each question.

Interesting:

A response is considered interesting if it presents information in an engaging manner. Do not mark the response as uninteresting just because the topic does not interest you, instead think about whether the person in the conversation would find it interesting.

No: the response is dull, and would not be considered interesting by anybody.

- *Hi there.*
- *Oh wow! That's cool!*

Somewhat: the response is somewhat interesting, but lacking in its content or delivery. Something that isn't dull but not particularly interesting.

- *Oh, I'm sorry. Well, I guess you'll just have to see if you can get the doctor to prescribe something for it.*
- *Nope. I'm not very good with cooking.*

Yes: the response is interesting in its content and delivery.

- *I have a feeling that if you can dodge a wrench you can dodge a ball.*
- *The Last Airbender, and Arrested Development are my favorites.*

Engaging:

A response is considered engaging if it can engage the user. This might be an inquisitive question or an interesting response that can be followed-up on.

No: the response is boring and does little to engage the user.

- *Hi there.*
- *Oh wow! That's cool!*

Somewhat: the response is not particularly engaging but still leaves room for follow-up.

- *My favourite colour is blue.*
- *Nope. I'm not very good with cooking.*

Yes: the response is actively engaging the user and trying to move forward the conversation.

- *I have a feeling that if you can dodge a wrench you can dodge a ball.*
- *What kind of shows do you like?*

Specific:

A response is considered generic if it's a boilerplate response that could work in a lot of different contexts. A response is specific if it is tailored to the specific conversation.

No: the response is generic and could be used in a lot of different contexts with some modification.

- *Hi there.*
- *Oh wow! That's cool!*

Somewhat: the response is somewhat specific, but has elements of a generic response. For example, it begins with a very generic acknowledgement of the user then asks a question.

- User: *Wooden toys, I carve them out by hand.* System: *That's awesome, how long have you been at it.*
- User: *It's going great, I just finished reading '1001 inventions that changed the world'* System: *That sounds interesting. How was it?*

Yes: the response is specific to the conversation.

- User: *I think I'm starting to feel sick actually* System: *I hope you feel better soon! If you don't mind me asking, why are you feeling sick?*
- User: *I think I like most foods. I really like steak* System: *Steak is good. I especially love it when it's well done.*

Relevant:

A response is considered relevant if it is on topic and a valid/reasonable continuation of the conversation. An irrelevant response is one that is completely off topic, or abruptly tries to change the topic of conversation.

No: the response is irrelevant. It is completely off topic or abruptly tries to change the topic of conversation.

- User: *I think I'm starting to feel sick actually* System: *My favorite movie is Parasite*
- User: *Can you play dodge ball?* System: *What kind of shows do you like?*

Somewhat: the response is a valid continuation of the conversation, but is somewhat generic or tangential to the main topic. For example, it begins with a generic acknowledgement of the topic then changes topics.

- User: *Have you tried seafood pancake?* System: *It's okay. I like beef bulgogi, but I'm not a huge fan of kimchi.*
- User: *It's going great, I just finished reading '1001 inventions that changed the world'* System: *That sounds interesting. My favorite movie is Parasite.*

Yes: the response is relevant, on topic and a valid continuation of the conversation.

- User: *lol... wanna join us for dodge ball?* System: *Sounds like fun! I'm in!*
- User: *I think I like most foods. I really like steak* System: *Steak is good. I especially love it when it's well done.*

Correct:

A response is considered correct if the system response is valid. It is incorrect if the response suggests that the system misunderstood what the user was asking.

No: the response is incorrect. The system completely misunderstood what the user was saying and the response doesn't make sense in the context of the conversation.

- User: *I think so but still get a lot of work to do.* System: *Therefore you are. Still how long? I don't know if I can.*
- User: *Have a meeting now. Need to go.* System: *Right now? No thanks but you can send me a donation. See you later.*

Somewhat: the response is somewhat incorrect. The response is not completely invalid, but it suggests the system misunderstood the context.

- User: *Kind of sunny every. But today is cloudy and cold.* System: *Many people prefer the Sun to the rain but they are both equally as important for the planet. Congratulations.*
- User: *What do you like most about the snow?* System: *The special effects are the best*

Yes: the response is correct and does not signify a misunderstanding of the conversation.

- User: *lol... wanna join us for dodge ball?* System: *Sounds like fun! I'm in!*
- User: *I think I like most foods. I really like steak* System: *Steak is good. I especially love it when it's well done.*

Semantically Appropriate:

A response is considered semantically appropriate if it makes sense in the context of the real world. A response is not semantically appropriate if it is presenting incorrect information or merging concepts that don't make sense.

No: the response is not semantically appropriate. It does not make sense in the context of the real world.

- User: *What do you like most about the snow?* System: *The special effects are the best*
- User: *I think so but still get a lot of work to do.* System: *Therefore you are. Still how long? I don't know if I can.*

Somewhat: the response is somewhat appropriate. The response makes sense, but it is strange and out of place.

- User: *Kind of sunny every. But today is cloudy and cold.* System: *Many people prefer the Sun to the rain but they are both equally as important for the planet. Congratulations.*
- User: *haha sounds like a video game thing* System: *I would have said the same thing about dodgeball.*

Yes: the response is semantically appropriate and makes sense.

- User: *I like dramas* System: *The Sopranos, Avatar: The Last Airbender, and Arrested Development are my favorites.*
- User: *I think I like most foods. I really like steak* System: *Steak is good. I especially love it when it's well done.*

Understandable:

A response is considered understandable if you can understand it's meaning. Even if it does not make sense in context or if the system misunderstood the user, if you can understand what the system is trying to say -- the response is understandable.

No: the response is not understandable. You do not know what the system is trying to say

- User: *I think so but still get a lot of work to do.* System: *Therefore you are. Still how long? I don't know if I can.*

Yes: the response is understandable.

- User: *I like dramas* System: *The Sopranos, Avatar: The Last Airbender, and Arrested Development are my favorites.*
- User: *I think I like most foods. I really like steak* System: *Steak is good. I especially love it when it's well done.*

Fluent:

A response is considered fluently written if it follows all the rules of English grammar.

No: the response is not fluently written.

- *Therefore you are. Still how long? I don't know if I can.*
- *OK let's do it. Pardon me? Why it shouldn't be?*

Somewhat: the response is somewhat fluent, but makes a minor mistake or is confusing.

- *I am 18 of your Earth years.*
- *Glad to hear it. What's new? Oh. Maybe you should get some rest?*

Yes: the response is written fluently.

- *I have a feeling that if you can dodge a wrench you can dodge a ball.*
- *The Last Airbender, and Arrested Development are my favorites.*

Coherent:

A conversation is considered coherent if the system's responses are relevant and semantically appropriate throughout a conversation. A coherent conversation maintains good flow, in that it does not abruptly change topics or lose track of the conversation.

No: the conversation is incoherent.

- The system changes topics abruptly.
- The system loses track of the conversation and at several points appears to misunderstand the user

Somewhat: the response is generally coherent, but makes mistakes in a few places

Yes: the system is always coherent and there is good conversation flow. The responses are all relevant and make sense.

Error Recovery:

Systems often make mistakes, and we'd like to measure whether they successfully recover from errors. At some point in the dialog a system might generate a bad response, and this question is concerned with whether it is able to recover from it in the next turn.

No: the system makes an error and never recovers from it, either changing the topic or continuing to produce incorrect outputs.

Somewhat: either the system somewhat recovers, it recovers from some errors but not all or the human is the one that recovers from errors.

Yes: the system successfully recovers from the error and the conversation continues successfully.

N/A: there were no errors for the system to recover from.

Consistent:

A system is consistent if it does not contradict itself throughout the conversation. An inconsistent system is one that contradicts something it said earlier.

No: the system repeatedly and obviously contradicts something it said earlier.

- System (turn 3): *I love steak. Especially when it's well done* System (turn 5): *Yes I love medium-rare steak the best*
- System (turn 2): *It's so sunny today* System turn (4): *It rained all day today. I had to carry an umbrella.*

Yes: the system is consistent and does not contradict itself.

Diverse:

A system is diverse if it produces responses in terms of content, style and structure. A non-diverse system is one that repeatedly generates generic responses or responses that follow a similar structure.

No: the system repeatedly generates very similar responses.

- The system generates three responses throughout the conversation that start with *That's awesome!*
- All of the systems responses are generic or dull.

Somewhat: the system is somewhat diverse in its responses, however there is a bit of repetition

Yes: the system is diverse in terms of content, style and sentence structure.

Topic Depth:

A system has topic depth if it discusses topics at a deep level, going into specifics rather than producing generic responses.

No: the system repeatedly generates generic responses that don't go in depth about a topic.

- When discussing carpentry, the system asks questions like *How long have you been doing it?* or *Do you enjoy working on it?*

Somewhat: the system goes into some depth regarding a topic. The responses are not generic but they don't show a solid understanding of the topic.

- When discussing carpentry, the system asks questions like *What kinds of things do you usually build?* or *Do you have a favorite type of wood?*

Yes: the system discusses topics in depth and generates engaging and interesting responses.

- When discussing food, the system generates responses like *I don't like my steaks well done, I feel that you lose a lot of flavor.* or *I like beef bulgogi, but I'm not a huge fan of kimchi*

Likeable:

A system is likeable if it displays positive attributes like politeness, engagingness, empathy and charisma.

No: the system is dull, impolite and not engaging. Either this system does not have a personality or it is not a pleasant one.

Somewhat: the system has a personality, but it is not particularly positive or negative.

Yes: the system has a strong personality and it is polite, engaging, empathetic and/or charismatic.

Understanding:

A system understands the user if most of the responses are relevant to the conversation and signifies an understanding of the user message. If the user makes a request, the system replies effectively to this request.

No: the system does not understand the user at all. It consistently produces irrelevant or generic responses, that suggest the user's responses are largely being ignored.

Somewhat: the system sometimes understands the user, however it frequently fails to

Yes: the system has generally understands the user and its responses are sufficiently conditioned on the user's responses

Flexible:

A system is considered flexible if it adapts the topic of conversation based on the user's responses and the user's interests. A rigid system would be one that tries to talk about the same topic, even if the user tries to change the topic or express disinterest.

No: the system does not change topics even after attempts from the user.

- The user says they're not interested in football, but the system keeps bringing up football
- The user brings up movies, but the system ignores it and keeps talking about the cooking

Somewhat: the system sometimes adapts to the user, but it is clear that it has limits. Either it does not understand the user properly, or it fails to discuss a specific topic

Yes: the system is flexible and generally adapts to the user. If the user wants to direct the conversation, the system lets it.

Informative:

A system is informative if throughout the dialog it presents engaging information, either interesting facts or information about itself.

No: the system is not informative and presents little to no information throughout the dialog

Somewhat: the system presents some information, however either the information is generic/boring or informative responses are infrequent

Yes: the system is informative and has multiple informative responses, with interesting and engaging information

Inquisitive:

A system is inquisitive if throughout the dialog it asks questions and tries to engage the user. These questions can be about the user or about something the user mentioned.

No: the system is not inquisitive and never asks any questions.

Somewhat: the system sometimes asks questions, however these questions are infrequent or generic

Yes: the system is inquisitive and asks many questions throughout the conversation.

Context (1/4)

0. User: Hi!
1. System: Hello! :D How are you?
2. User: Do you like Java, C++ Or Python?

Response

System: C++. :)

Instructions: Please answer the following questions about the **response** shown on the left. Read the conversation thoroughly and please consult the instructions for each question if needed. If a question does not apply for the response, you may select N/A and briefly justify.

1. To the average person, is the response **interesting?** ([show instructions](#))

- ☐ No
☐ Somewhat
☐ Yes
☐ N/A

If you chose N/A. Please justify.

2. Is the response **engaging?** ([show instructions](#))

- ☐ No
☐ Somewhat
☐ Yes
☐ N/A

If you chose N/A. Please justify.

3. Is the response **generic or **specific** to the conversation?** ([show instructions](#))

- ☐ Generic
☐ Somewhat generic
☐ Specific
☐ N/A

If you chose N/A. Please justify.

4. Is the response **relevant to the conversation?** ([show instructions](#))

- ☐ No
☐ Somewhat
☐ Yes
☐ N/A